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个人简介/研究概述

长期从事光场调控、计算显微成像与光谱量测技术研究, 聚焦超分辨三维成像、红外光热化学成像、光学衍射层析和微纳结构定量表征。近年来以第一/共同第一作者及通信作者在 *Nature Communications*, *Light: Science & Applications*, *Optica*, *Advanced Photonics* 等期刊发表论文, 并主持国家自然科学基金青年项目、国家重点研发计划青年科学家课题子课题等项目。

研究方向: 面向先进制造的前沿光学技术研究及应用

[光场调控与计算成像] 以光场调控为核心, 发展高时空分辨、高信息内涵的先进光学成像、光谱与量测方法。

[激光制造与光学表征] 突破超快非线性光学、三维高分辨制造及折射率、形貌、组成等定量表征关键技术。

[系统集成与交叉应用] 面向激光极端制造、新型材料与微纳器件, 推动原创光学测量/制造技术与先进制造应用。

工作经历

博士后研究员	2023.10 - 至今
波士顿大学电子与计算机工程系 Ji-xin Cheng 教授课题组	波士顿
特聘副研究员	2021.7 - 2023.10
北京大学物理学院施可彬教授课题组	北京
博士后	2019.7 - 2021.6
北京大学物理学院施可彬教授课题组	北京

教育背景

北京大学	博士-光学 导师: 施可彬教授	2014 - 2019
南开大学	本科-物理学	2010 - 2014

代表性论文

- [超分辨化学成像] Dashan Dong[#], Danchen Jia, Xinyan Teng, Jianpeng Ao, George Abu-Aqil, Biwen Gao, Meng Zhang, Qing Xia, Ji-Xin Cheng[✉]. “ Chem-SIM: SIM-resolved chemical imaging via wide-field mid-infrared photothermal modulation of fluorescence ”. *Nature Communications*, (2026).
- [单颗粒材料成像] Lige Liu[#], Dashan Dong[#], Zhiwei Long, Wanxue Wei, Chang Sun, Wei Liu, Xiaoshuai Huang, Liangyi Chen, Haizheng Zhong[✉], and Kebin Shi[✉]. “ Single particle fluorescence imaging of perovskite nanocrystal crystallization for illustrating coupled nucleation-and-growth ”. *Nature Communications*, 16.5664, (2025).
- [计算偏振/双折射成像] Shuqi Mu[#], Yingting Shi, Yintong Song, Wei Liu, Wanxue Wei, Qihuang Gong, Dashan Dong[✉], and Kebin Shi. “ Multislice computational model for birefringent scattering ”. *Optica*, 10.1, (2023).
- [微纳结构三维表征] Shuqi Mu[#], Yu Yang, Juhao Li, Dashan Dong[✉], Ruijun Lan[✉], Kebin Shi. “ Three dimension refractive index characterization for photonic waveguides ”. *Journal of Lightwave Technology*, 40.8, (2022).
- [活细胞三维超分辨成像] Dashan Dong[#], Xiaoshuai Huang[#], Liuju Li[#], Heng Mao, Yanquan Mo, Guangyi Zhang, Zhe Zhang, Jaiyu Shen, Wei Liu, Zeming Wu, Guanghui Liu, Yanmei Liu, Hong Yang, Qihuang Gong, Kebin Shi[✉], and Liangyi Chen[✉]. “ Super-resolution fluorescence-assisted diffraction computational tomography reveals the three-dimensional landscape of the cellular organelle interactome ”. *Light: Science & Applications*, 9.11, (2020).

- (1) Xu Zhang[#], Yonghao Chen, Shulei Zhou, Xiaoshuai Huang, Yujie Sun, Hong Yang, **Dashan Dong**[✉], Yue Wang[✉], Kebin Shi[✉]. “ Common-path Dark-full-field Optical Coherence Tomography with Asymmetric Detection and Spatial Illumination [✉](#) .” *Light: Science & Applications*, Under Review.
- (2) Danchen Jia[#], **Dashan Dong**[#], Tongyu Li, Haonan Zong, Jiabei Zhu, Xinyan Teng, Lei Tian[✉], Ji-Xin Cheng[✉]. “ [Mid-Infrared Photothermal Relaxation Intensity Diffraction Tomography for Video-rate Volumetric Chemical Imaging \[✉\]\(#\)](#) .” *Optics Express*, 34.9, (2026).
- (3) George Abu-Aqi[#], **Dashan Dong**[#], Jiaze Yin[#], Jianpeng Ao, Hongjian He, Guangrui Ding, Qing Xia, Ji-Xin Cheng[✉]. “ [Bond-Selective Imaging via Vibrational Relaxation Encoded Fluorescence \[✉\]\(#\)](#) .” *The Journal of Physical Chemistry Letters*, 16, (2025).
- (4) Jia Danchen[#], Huang Steven[#], Tulegenov Dias, **Dashan Dong**, Shvets Gennady[✉], Cheng Ji-Xin[✉]. “ [Metasurface-enhanced infrared photothermal microscopy towards ultrasensitive chemical analysis \[✉\]\(#\)](#) .” *Advanced Photonics*, 7.5, (2025).
- (5) Qian Chen[#], Wen Gou[#], Wenqing Lu[#], Jie Li, Yuhong Wei, Haoyu Li, Chengyu Wang, Wei You, Zhengqian Li, **Dashan Dong**, Xiuli Bi, Bin Xiao[✉], Liangyi Chen[✉], Kebin Shi[✉], Junchao Fan[✉], Xiaoshuai Huang[✉]. “ [Fast, three-dimensional, live-cell super-resolution imaging with multiplane structured illumination microscopy \[✉\]\(#\)](#) .” *Nature Photonics*, (2025).
- (6) Jiaze Yin[#], Christian Pfluegl, Chu C. Teng, Rylie Bolarinho, Guo Chen, Xinrui Gong, **Dashan Dong**, Daryoosh Vakhshoori, Ji-Xin Cheng[✉]. “ [Mid-infrared Energy Deposition Spectroscopy \[✉\]\(#\)](#) .” *Physical Review Letters*, 134.9, (2025).
- (7) Yue Wang[#], Jingrui Gong, Ning Xu, Shaohui Yan, **Dashan Dong**, Kebin Shi[✉]. “ [Large Field of View and Isotropic Light Sheet Microscopy with Aberration-Free Tunable Foci \[✉\]\(#\)](#) .” *Laser & Photonics Reviews*, 19.3, (2025).
- (8) Wenkai Yang[#], Zijian Wang, Jian Xu, **Dashan Dong**, Guiyuan Cao, Han Lin, Baohua Jia, Lige Liu[✉], Kebin Shi[✉]. “ [Ultra-compact computational spectroscopy with a detour-phased planar lens \[✉\]\(#\)](#) .” *Light: Advanced Manufacturing*, 5.44, (2024).
- (9) Ziheng Ji[#], Wentao Yu, **Dashan Dong**, Hong Yang, Kaihui Liu, Yun-Feng Xiao, Qihuang Gong, Qinghai Song[✉], Kebin Shi[✉]. “ [High spatial resolution collinear chiral sum-frequency generation microscopy \[✉\]\(#\)](#) .” *Advanced Photonics Nexus*, 3.2, (2024).
- (10) Yue Wang[#], **Dashan Dong**, Wenkai Yang, Renxi He, Ming Lei, and Kebin Shi[✉]. “ [Reflective ultrathin light-sheet microscopy with isotropic 3D resolutions \[✉\]\(#\)](#) .” *Photonics Research*, 12.2, (2024).
- (11) Peng Liu[#], Jing Shi[#], Danli Sheng[#], Wenqing Lu, Jie Guo, Lei Gao, Xiaoqing Wang, Shaofeng Wu, Yanwen Feng, **Dashan Dong**, Xiaoshuai Huang[✉], and Hongyun Tang[✉]. “ [Mitophrogenesis, a form of mitochondria-specific ectocytosis, regulates sperm mitochondrial quantity and fertility \[✉\]\(#\)](#) .” *Nature Cell Biology*, 25, (2023).
- (12) 魏婉雪[#], 何苗, 徐坚, 董大山[✉]. “ [一种大视野快速光学衍射层析成像技术\[✉\]\(#\)](#) .” *中国激光*, 50.3, (2023).
- (13) Wenkai Yang[#], Lige Liu[✉], **Dashan Dong**, Yunan Gao, Hong Yang, Qihuang Gong, and Kebin Shi[✉]. “ [In situ three-dimensional observation of perovskite crystallization revealed by two-photon fluorescence imaging \[✉\]\(#\)](#) .” *Advanced Optical Materials*, 10.13, (2022).
- (14) Wenkai Yang[#], Lige Liu[#], **Dashan Dong**, Xin Zhang, Han Lin, Yunkun Wang, Hong Yang, Yunan Gao, Haizheng Zhong, Baohua Jia, and Kebin Shi[✉]. “ [Detour-phased perovskite ultrathin planar lens using direct femtosecond laser writing \[✉\]\(#\)](#) .” *Photonics Research*, 10.12, (2022).
- (15) Runfeng Li[#], Ruijun Lan[✉], **Dashan Dong**, Hong Yang, and Kebin Shi[✉]. “ [Bessel beam coherent anti-Stokes Raman scattering spectroscopy for turbulent flow diagnosis \[✉\]\(#\)](#) .” *Applied Spectroscopy*, (2022).
- (16) 穆书奇[#], 董大山, 施可彬[✉]. “ [无标记光学成像技术\[✉\]\(#\)](#) .” *激光与光电子学进展*, 59.12, (2022).
- (17) 李润丰[#], 董大山, 施可彬[✉]. “ [光场调控在相干拉曼散射光谱与成像中的应用 \(特邀\)\[✉\]\(#\)](#) .” *光子学报*, 50.1, (2022).
- (18) Wei Liu[#], **Dashan Dong**, Hong Yang, Qihuang Gong, and Kebin Shi[✉]. “ [Robust and high-speed rotation control in optical tweezers by using polarization synthesis based on heterodyne interference \[✉\]\(#\)](#) .” *Opto-Electronic Advances*, 3.8, (2020).

专利

(1) <i>System and Method for Super-Resolution Chemical Imaging Via Photothermal</i>	PCT/US26/24003, (2026)
(2) 一种双模态显微成像系统和方法	PCT/CN2021/071393, (2021)
(3) 一种基于缺失型反射镜的离轴全息合束装置及其方法	2023110715739, (2023)
(4) 一种双模态显微成像系统及其成像方法	ZL202110946936.3, (2022)
(5) 基于微棱镜的反射式轴向光片荧光显微成像装置及方法	ZL 202110817118.3, (2022)
(6) 一种基于贝塞尔 CARS 的湍流光谱测量系统及其测量方法	ZL202110667298.1, (2022)
(7) 一种双模态显微成像系统和方法	ZL202010059510.1, (2022)
(8) 基于光学外差干涉法的动态柱矢量光场产生装置及其方法	ZL202010493775.2, (2021)
(9) 一种基于 GB-STED 的深层超分辨激光直写系统及其实现方法	ZL202010069141.4, (2020)

科研项目

[课题负责人] 国家重点研发计划下设青年科学家课题 (2022YFC3401103)	2022
活细胞超分辨率三维全景成像与解析技术体系研发课题：无标记成像与大数据机器学习	384 万元
[主持] 国家自然科学基金青年科学基金项目 (12004013)	2021
用于活细胞成像的反射式光学衍射层析成像及其应用	24 万元
[共同负责人] 北京大学临床医学 +X 青年专项	2022
基于无标记成像探究脂滴在卵不对称分裂中的作用	10 万元
[主持] 中国博士后科学基金第 68 批面上资助 (2020M680220)	2021
超分辨荧光辅助的光学衍射层析成像及其应用	8 万元
[参与] 国家自然科学基金面上项目 (12174010)	2022
微腔混沌动力学的超高速光学成像	62 万元

学术服务

[审稿人] <i>Photonics Research, Optics Letters, Optics Express, Advanced Science, APL Photonics</i> , 中国激光, 光学学报	
[程序委员会委员] 国际计算成像会议 (CSOE-CITA), 专题 9: 生物医学与计算成像	2022, 2023
[青年编委] 《中国激光》杂志	2022, 2023
[学生组织服务] 北京大学 SPIE 学生分会	主席 (2017), 副主席 (2016)

荣誉奖项

[2022] 人工微结构和介观物理国家重点实验室甘子钊优秀博士后奖	北京大学
[2022] 现代光学所重要科研进展	北京大学
[2017] 五四奖学金	北京大学
[2017] 优秀科研单项奖	北京大学
[2013] 合展励学金	中国科技馆发展基金会
[2012] 国家励志奖学金	南开大学
[2012] 第三届中国大学生物理学术竞赛-团队一等奖	
[2012] 三好学生	南开大学
[2011] 优秀学生基础学科奖学金	南开大学